

Amazon SNS Standard Topic with Email Subscription and S3 Publisher

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon SNS Standard Topic** was created and used to send notifications to an email subscriber. First, the topic was created with default settings, and then an email subscription was added to it. Because email subscriptions require confirmation, the subscriber initially stayed in a pending state until the confirmation link was clicked. After that, a test message was published directly from the SNS topic, and the email inbox received the notification successfully. The lab then connected **Amazon S3** as a publisher by enabling bucket event notifications so that every object upload in the bucket triggered SNS. Finally, uploading a file to the S3 bucket sent a notification to the confirmed email subscriber, proving the full publisher → SNS → subscriber flow.

Understanding Amazon SNS

Amazon SNS is a publish-subscribe messaging service that lets one publisher send a message to many subscribers at the same time. It exists so applications and AWS services can broadcast notifications without needing to know who the final receivers are. In this lab, SNS acted as the middle layer between **Amazon S3** and the email inbox. That makes it useful for alerts, event notifications, and fan-out messaging where one event needs to reach multiple endpoints.

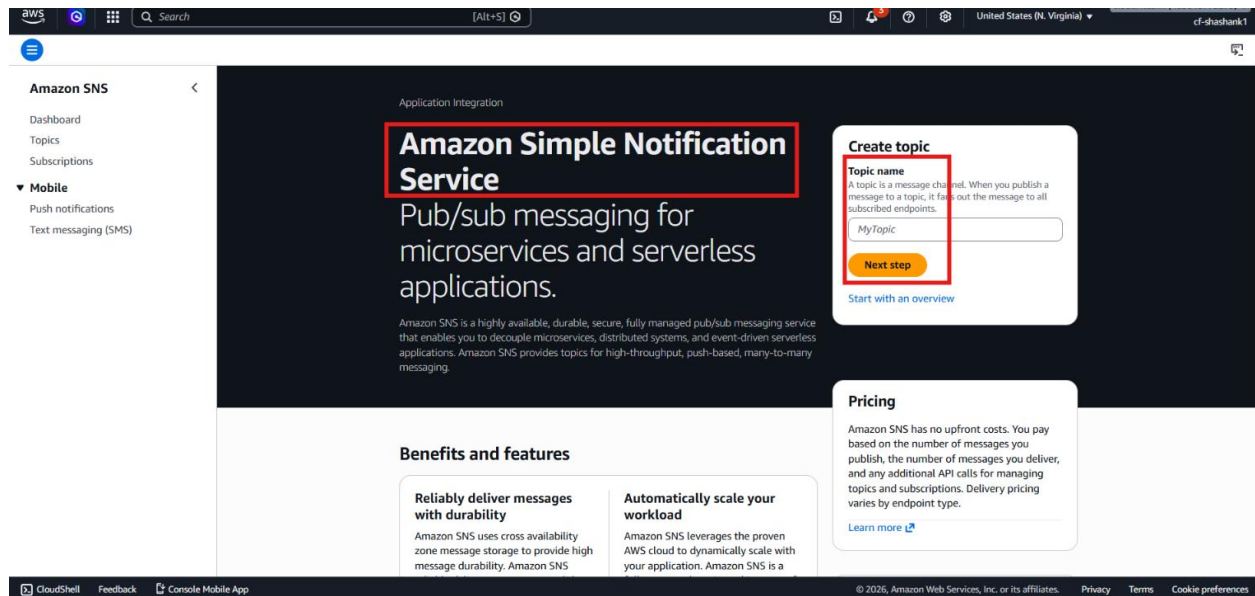
Example:

A new file is uploaded into an S3 bucket named project-assets. S3 sends the event to an SNS topic, and that topic delivers the message to an email subscriber. If later you add more subscribers like Lambda, HTTP, or SQS, the same single S3 event can notify all of them.

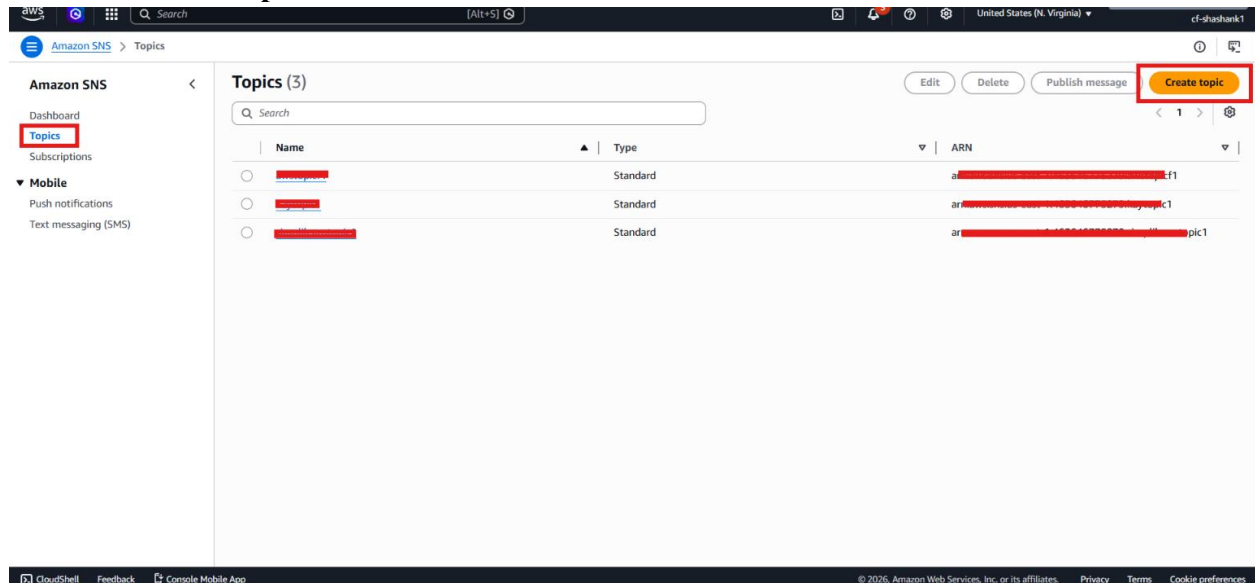
Lab Steps

Step 1 – Create the SNS Standard Topic

- Sign in to the AWS Management Console.
- Open **Amazon SNS**.



- Click **Create topic**.



- Select **Standard** topic type.
- Enter the topic name: NewTopic
- Add a display name if required but I will give it as CloudFreeeks.
- Keep the default settings.

Create topic

Details

Type [Info](#)
Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ **Standard**

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

NewTopic

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional [Info](#)
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

CloudFreaks

Maximum 100 characters.

► **Encryption - optional**
Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

► **Access policy - optional** [Info](#)
This policy defines who can access your topic. By default, only the topic owner can publish or subscribe to the topic.

- Click **Create topic**.

Step 2 – Create the Email Subscription

- Open the newly created SNS topic.
- Click **Create subscription**.

Amazon SNS

Dashboard
Topics
Subscriptions

▼ **Mobile**
Push notifications
Text messaging (SMS)

Topic NewTopic created successfully.
You can create subscriptions and send messages to them from this topic.

NewTopic

[Edit](#) [Delete](#) [Publish message](#)

Details

Name NewTopic	ARN NewTopic	Display name CloudFreaks	Type Standard
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Topic owner
463646775279

[Subscriptions](#) [Access policy](#) [Delivery policy \(HTTP/S\)](#) [Delivery status logging](#) [Encryption](#) [Tags](#) [Integrations](#)

Subscriptions (0)

[Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

ID	Endpoint	Status	Protocol
No subscriptions found You don't have any subscriptions to this topic.			

[Create subscription](#)

- Select **Email** as the protocol.
- Enter the email address as the endpoint.
- Click **Create subscription**.

Amazon SNS > Subscriptions > Create subscription

Details

Topic ARN
[Redacted] NewTopic

Protocol
The type of endpoint to subscribe
Email

Endpoint
An email address that can receive notifications from Amazon SNS.
[Redacted]@gmail.com

After your subscription is created, you must confirm it. [Info](#)

Subscription filter policy - optional [Info](#)
This policy filters the messages that a subscriber receives.

Redrive policy (dead-letter queue) - optional [Info](#)
Send undeliverable messages to a dead-letter queue.

[Cancel](#) [Create subscription](#)

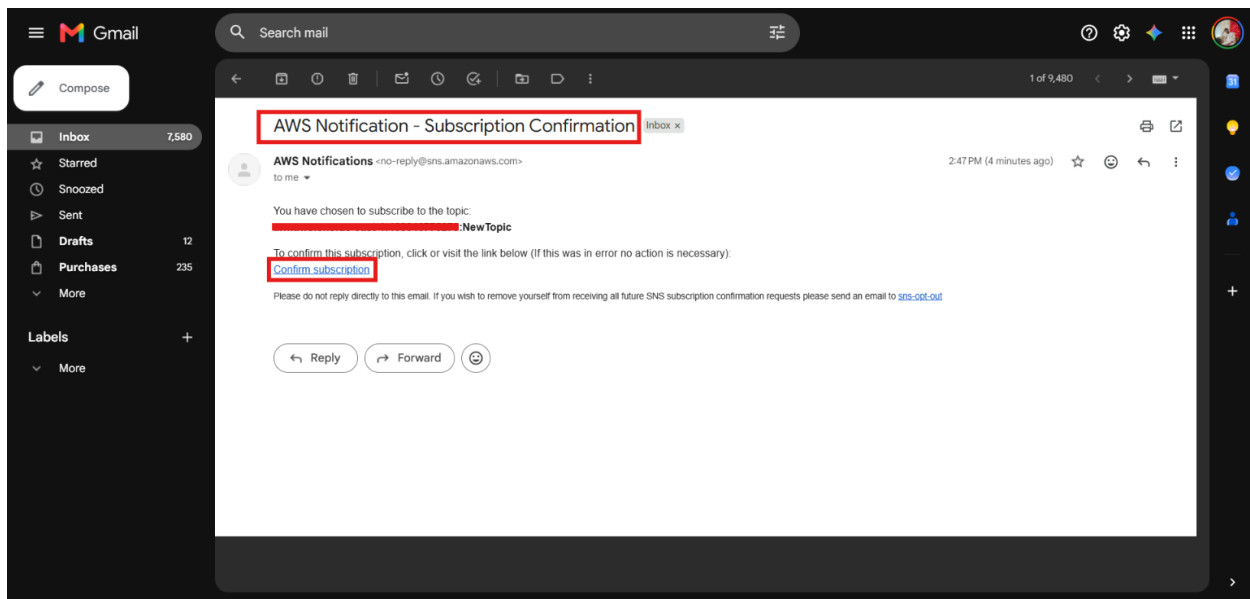
- Wait for the subscription status to become pending.

Amazon SNS > Subscriptions

Subscriptions (4)

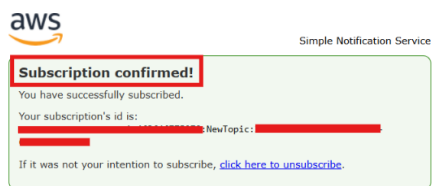
ID	Endpoint	Status	Protocol	Topic
[Redacted]	[Redacted]	Confirmed	EMAIL	[Redacted]
[Redacted]	[Redacted]	Confirmed	EMAIL	[Redacted]
[Redacted]	[Redacted]	Confirmed	EMAIL	[Redacted]
Pending confirmation	[Redacted]@gmail.com	Pending confirmation	EMAIL	NewTopic

- Open the email inbox and confirm the subscription by clicking the confirmation link.

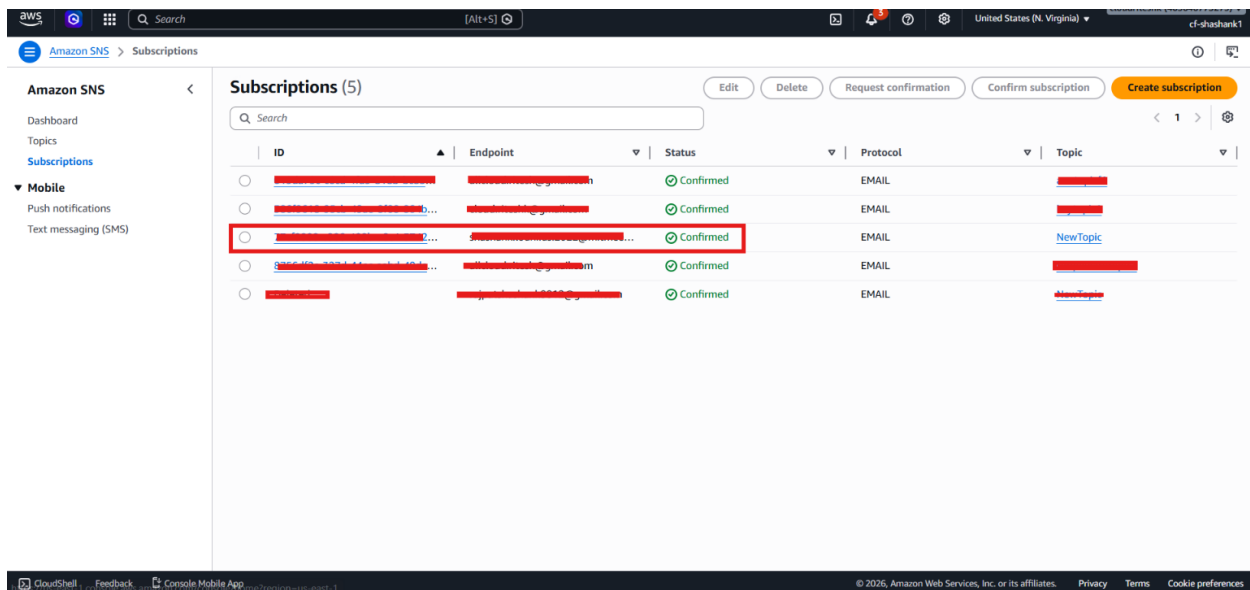


Step 3 – Verify the Subscription

- Return to the SNS console after confirming the email link.



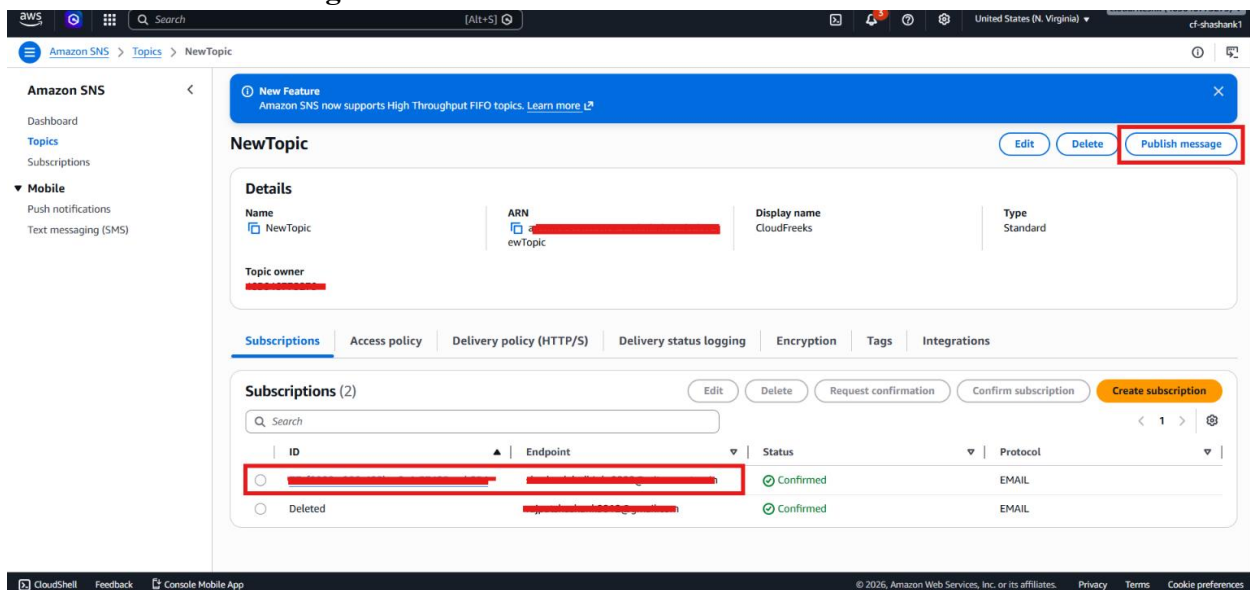
- Refresh the subscription list.
- Confirm that the email subscriber status is now confirmed.



- Note that a single topic can have multiple subscribers if needed.

Step 4 – Publish a Test Message

- Open the SNS topic.
- Click **Publish message**.



- Enter a subject: Hello

Amazon SNS > Topics > NewTopic > Publish message

New Feature
Amazon SNS now supports High Throughput FIFO topics. [Learn more](#)

Publish message to topic

Message details

Topic ARN
arn:aws:sns:us-east-1:123456789012:myTopic

Subject - optional
Hello

Maximum 100 printable ASCII characters

Message group ID - optional
An optional token for SQS fan queue usage. Propagated only to standard SQS queue subscriptions.
Enter message group ID

Maximum 128 characters. Can include alphanumeric characters and punctuation.

Time to Live (TTL) - optional [info](#)
This setting applies only to mobile application endpoints. The number of seconds that the push notification service has to deliver the message to the endpoint.

Message body

Message structure

☒ Identical payload for all delivery protocols.
The same payload is sent to endpoints subscribed to the topic, regardless of their delivery protocol.

☐ Custom payload for each delivery protocol.
Different payloads are sent to endpoints subscribed to the topic, based on their delivery protocol.

- Enter the message body: Hello from CloudFreeks
- Click **Publish message**.

Amazon SNS > Topics > NewTopic > Publish message

Message body

Message structure

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Message body to send to the endpoint

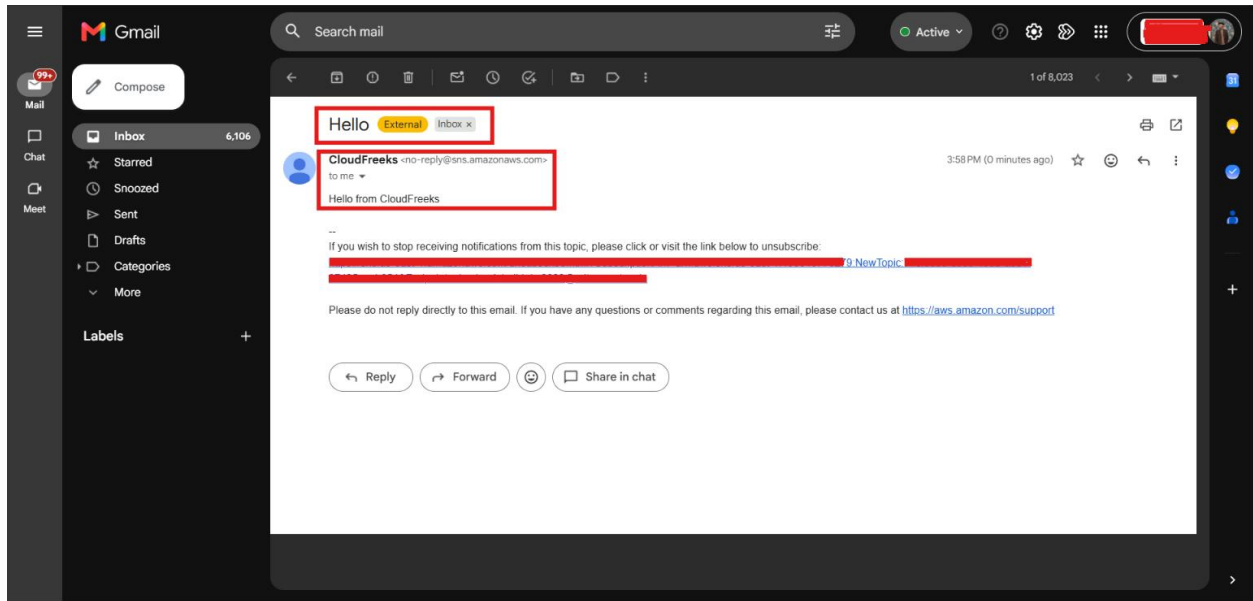
1 Hello from CloudFreeks

1:23 Text

Message attributes [info](#)
Message attributes let you provide structured metadata items (such as timestamps, geospatial data, signatures, and identifiers) for the message.

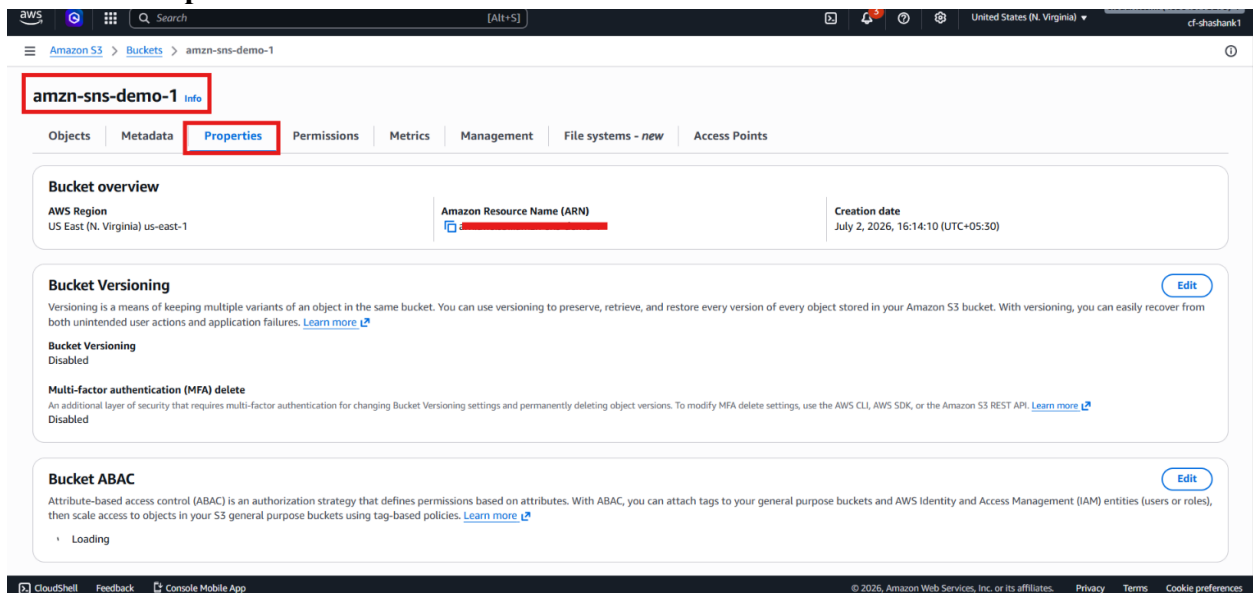
Type	Name	Value	
Select attribute type	Enter attribute name	value or ["value1", "value2"]	Remove

- Check the email inbox.
- Confirm that the SNS notification arrived successfully.



Step 5 – Configure Amazon S3 as a Publisher

- Open Amazon S3.
- Select any bucket or create one.
- Go to **Properties**.



- Open Event notifications.
- Click **Create event notification**.

Event notifications (0)

Send a notification when specific events occur in your bucket. [Learn more](#)

No event notifications

Choose Create event notification to be notified when a specific event occurs.

Create event notification

Amazon EventBridge

For additional capabilities, use Amazon EventBridge to build event-driven applications at scale using S3 event notifications. [Learn more](#) or [see EventBridge pricing](#)

Send notifications to Amazon EventBridge for all events in this bucket

Off

Transfer acceleration

Use an accelerated endpoint for faster data transfers. [Learn more](#)

Transfer acceleration

Disabled

Object Lock

Store objects using a write-once-read-many (WORM) model to help you prevent objects from being deleted or overwritten for a fixed amount of time or indefinitely. Object Lock works only in versioned buckets. [Learn more](#)

Object Lock

Disabled

- Enter a notification name: New Upload
- Select the event type **All Object created**.

General configuration

Event name

New Upload

Event name can contain up to 255 characters.

Prefix - optional

Limit the notifications to objects with key starting with specified characters.

images/

Suffix - optional

Limit the notifications to objects with key ending with specified characters.

jpg

Event types

Specify at least one event for which you want to receive notifications. For each group, you can choose an event type for all events, or you can choose one or more individual events.

Object creation

☒ All object create events
s3:ObjectCreated:*

☐ Put
s3:ObjectCreated:Put

☐ Post
s3:ObjectCreated:Post

☐ Copy
s3:ObjectCreated:Copy

☐ Multipart upload completed

- Choose **SNS topic** as the destination.
- Select the SNS topic you created.
- Save the changes.

Destination

Before Amazon S3 can publish messages to a destination, you must grant the Amazon S3 principal the necessary permissions to call the relevant API to publish messages to an SNS topic, an SQS queue, or a Lambda function. [Learn more](#)

Destination
Choose a destination to publish the event. [Learn more](#)

☐ Lambda function
Run a Lambda function script based on S3 events.

☒ **SNS topic**
Route messages to systems for parallel processing or directly to people.

☐ SQS queue
Send notifications to an SQS queue to be read by a server.

Specify SNS topic

☒ Choose from your SNS topics

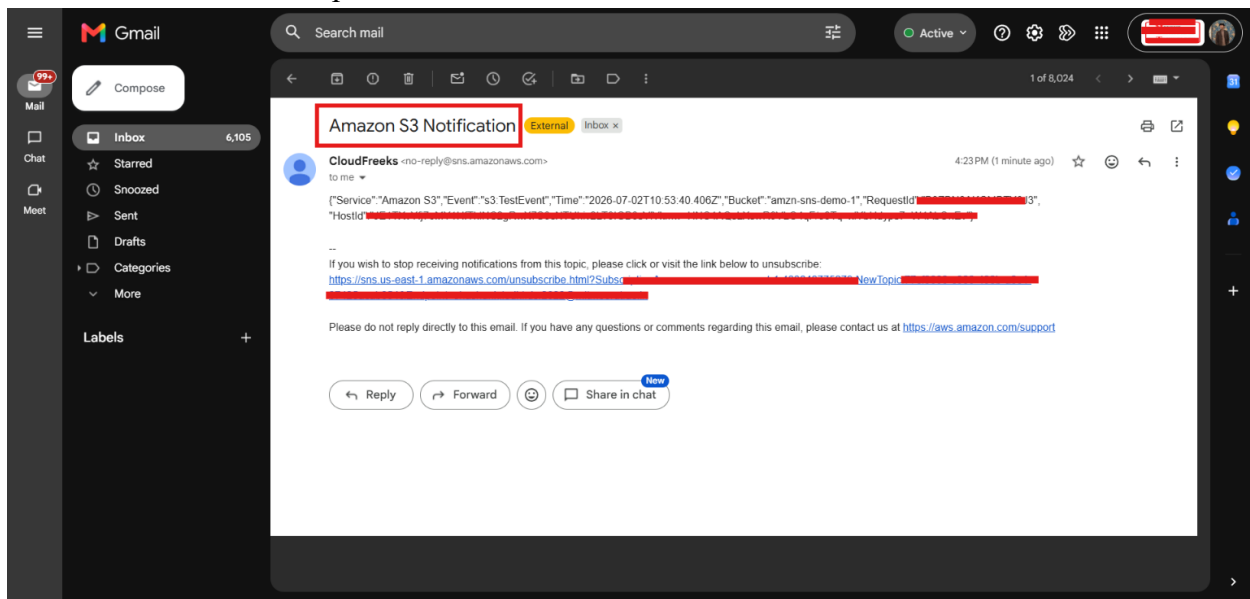
☐ Enter SNS topic ARN

SNS topic
NewTopic

[Cancel](#) [Save changes](#)

Step 6 – Test the S3 to SNS Flow

- Upload a file to the S3 bucket.
- Wait for the upload event to trigger.
- Check the email inbox.
- Confirm that the S3 upload notification was received.



- Verify that the message contains bucket and object details.

Verification

- An **Amazon SNS Standard Topic** was created successfully.
- An email subscriber was added and confirmed.
- A test message was published from SNS and received by email.

- **Amazon S3** was configured as a publisher using event notifications.
- Uploading a file to the bucket triggered an SNS notification.
- The complete publisher → SNS → subscriber flow worked correctly.